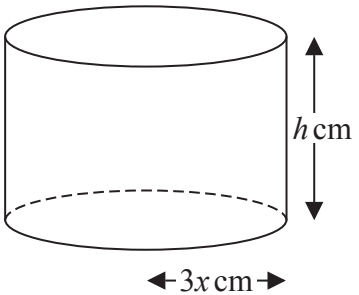
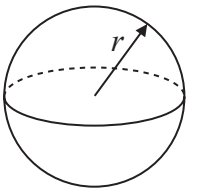
17 The diagram shows a solid metal cylinder.



Volume of sphere $= \frac{4}{3}\pi r^3$



The cylinder has base radius $3x$ cm and height h cm.

The metal cylinder is melted.
All the metal is then used to make 270 spheres.

Each sphere has a radius of $\frac{1}{2}x$ cm.

Find an expression, in its simplest form, for h in terms of x.

Cylinder volume = $\pi(3x)^2h = 9\pi x^2h$.

1 mark
Both volume formulae

One sphere = $\frac{4}{3}\pi(\frac{1}{2}x)^3 = \frac{\pi x^3}{6}$

270 spheres = $270 \times \pi x^3 \div 6 = 45\pi x^3$.

$9\pi x^2h = 45\pi x^3$

$\div(9\pi x^2)$ $\div(9\pi x^2)$

1 mark
Equal total volumes

$h = 5x$

1 mark
h = 5x

(Total for Question 17 is 3 marks)